

Practitioner Portfolio: Kraken BTC/USD Systematic Round-Trip Accumulator

Self-built, self-funded, live Dec 2, 2024 to Jan 21, 2025

What I Built

A reactive, post-fill paired-order strategy for BTC/USD on Kraken, written in Python. Mechanics:

- On each buy fill at price P, the bot places a paired sell limit at P + minimum profit delta (the smallest increment that books a profit after fees) and re-establishes a buy at P.
- Sells are placed only at prices that book a structural profit on the round trip. By design, every successfully completed buy-then-sell pair is positive expected value.
- The strategy operates within Kraken's max-open-orders cap. When the cap or available capital is hit, the deepest underwater positions are abandoned via market sell to free capacity for the active part of the strategy.
- Pure-passive by design. 99.99% of 336,644 fills were limit orders. The 28 market-order fills are forced bag-cuts triggered when capacity or capital ran out.

Operational objective alongside the strategy itself: scale 30-day notional to qualify for Kraken's \$10M maker-fee tier, where maker fees drop to zero. Below that threshold the strategy is structurally unprofitable at any practical delta.

What I Executed Operationally

- Capital deployed:** about \$21,971. Starting balance of about \$11,982 at Dec 2; deposited about \$9,989 across Dec 2 to 6 to support scaled lot sizes.
- 50 days of live trading at scale**, 336,644 fills, **\$16.6M total notional** transacted on a single pair.
- Hit the \$10M tier on Dec 25 — maker fees dropped to zero.** Ran at 0 bps from Dec 27 to Jan 9 (14 days). Slipped one tier on Jan 10; ran the remaining 11 days at 2 bps.
- Capacity-relief logic exercised 28 times.** When max-open-orders and capital constraints were both hit, deepest bag positions were market-sold. \$283k of total taker volume, \$357 of taker fees.

Honest PnL As Run

Metric	Full run	Free-tier window only
Period	Dec 2 to Jan 21	Dec 27 to Jan 9
Net PnL	-\$11,131	-\$1,238
Return on capital deployed	-51%	-10% in 14 days
Max drawdown	-51%	-27%
Total fees paid	\$4,285	\$68

PnL Decomposition: What Actually Happened

Component	Value	Notes
Limit-flow round-trip engine	about +\$6,000 latent value	83.88 BTC bought, 80.87 sold via limit. +3.01 BTC net inventory @ avg \$99,012. Closing price \$107,050.
Forced bag-cuts (capacity-relief)	about -\$22,000 forfeited	2.91 BTC sold @ avg \$97,163 vs \$107,050 close. Same BTC worth \$312k vs \$283k received.
Fees	-\$4,285	Maker fees at 0 bps (free tier) and 2 bps (standard tier).
Inventory carry and timing	residual ~\$9k (negative)	Drawdown carry on open bag positions.

The round-trip engine performed as designed across this window. Substantially all of the realized loss came from the capacity-relief mechanism: forced exits at average prices about 6% below where patient holding would have closed. The fix is not to the round-trip logic; it is to the capacity and capital-management layer above it.

Counterfactual That Frames the Risk Honestly

	Actual	Counterfactual: bag held to Jan 21 (no forced cuts)
End equity	\$10,840	\$39,141
Net PnL	-\$11,131	+\$17,169
Delta	n/a	+\$28,301

The counterfactual works because BTC recovered from its mid-Jan low to \$102k by Jan 28. In a market where BTC continued falling for another month, "with more capital" would have meant larger losses, not smaller ones. The right takeaway is that **the strategy must be capitalized to its drawdown distribution**, not "add more capital."

What I Got Right

- **Strategy design at the round-trip level.** Aggregate limit-flow value creation matched the design intent. Every paired sell that filled booked a structural profit by construction; the round-trip engine accumulated about \$6k of latent value over the window.
- **Operational execution.** Deliberate ramp from 0.0001 BTC to 0.0005 BTC lots (approximately \$8 to \$50 in USD at prevailing prices), stable parameters at scale, capital additions timed to support tier qualification, clean shutdown. Lot sizes were fixed in BTC - their USD value varied with BTC price throughout the run.
- **Tier qualification mechanics.** Calibrated lot sizes and pace to cross the \$10M / 30-day rolling volume threshold on the predicted date.
- **Scope discipline.** Single pair, single venue, no leverage, 99.99% passive maker, no parameter changes mid-run.
- **Capacity-relief logic worked as a fail-safe.** The bot did not blow up. It cut the bag to keep operating, exactly as designed when caps were hit.

- **Logging.** Full ledger and trade-level export preserved. Every fill, fee, order, and balance change reconstructable after the fact, enabling the decomposition above.

What I Got Wrong

- **Capital sizing was set by what I had, not by what the strategy needs.** No formal drawdown-tolerance budget was computed before deployment. A simple rolling-window analysis on historical BTC returns at this lot size would have shown the bag could not survive a typical 15 to 20% correction without cuts.
- **Capacity-relief was rule-based, not regime-aware.** The bot cut whenever caps were hit, irrespective of where the bag stood relative to its historical cost basis or whether the drawdown showed signs of stabilizing.
- **No hedging leg.** No correlated short (perp futures) to free capital tied up in inventory and extend drawdown tolerance per dollar.
- **Optimized fee tier when capital was the binding constraint.** Fee tier was a real prize, but \$4,761 of fees was a fraction of the \$18k forfeited via cuts. Capital headroom would have paid back faster than fee savings.
- **No live MTM, drawdown, or capacity dashboard.** Cut decisions were made in the dark. No real-time view of where the bag stood relative to price, capital, or order-slot ceiling at the moment of decision.

What I Tested and What the Data Said

After the post-mortem, I formed a hypothesis that a perp futures hedge would solve the capacity-relief problem by providing cash flow during BTC drawdowns. I built a synthetic backtest applying a hedge to the actual trade tape at hedge ratios of 0.5, 0.75, and 1.0.

The data falsified the hypothesis in this regime. The hedge generated about \$2,300 of cash buffer during BTC's -15% drawdown from Dec 17 to Jan 13, against actual cut volume of \$282,730 in the same period. The hedge was less than 1% of the cut volume, far short of the magnitude needed to prevent cuts. Additionally, BTC's overall +34% rally during the run cost the perp short about \$1,300 in directional drag at $h=1.0$.

The honest conclusion: the perp hedge is a regime-dependent tail-risk overlay, not a substitute for adequate capital. It is most useful at lower hedge ratios ($h=0.5$) where it captures most of the tail benefit at half the directional drag. The actual fix for cuts is a combination of capital sizing to the drawdown distribution and range-adaptive position sizing.

What I'm Working On Next

1. **Capital sized to the drawdown distribution.** Rolling-window historical analysis across 6-week windows shows about \$50k unhedged or \$30k with $h=0.5$ hedge covers 95% of recent windows; tail-adjusted (including 2018 / 2022 / Mar 2020) pushes that to about \$100k unhedged.
2. **Dynamic lot sizing for fee tier maintenance.** Lot size - how much BTC to buy per fill - was fixed at 0.0005 BTC throughout the run (approximately \$47-\$54 in USD as BTC price moved). The fix makes it respond to daily fill count instead. The objective is simple: keep 30-day rolling notional above the \$10M free-tier threshold regardless of how active the market is. On quiet days with few fills, the lot rises so each fill contributes more notional. On busy days with many fills, the lot drops toward the minimum. The signal driving this is fill frequency, not price movement. At the target fill

rate of 281 fills per hour, the daily notional target of \$333k implies a base lot of \$72 - the free tier floor. Dynamic lot sizing keeps the strategy on the free tier continuously rather than by accident.

3. **Profit-target-driven dynamic delta sizing.** The primary revenue lever is not delta directly - it is P^* , the target profit per round trip in dollars. Delta is a derived output: $\text{delta} = P^* / \text{lot_btc}$. This matters because lot size changes with fill count to maintain the free tier, and if delta were held fixed while lot size changes, the revenue-to-lot-size ratio would drift. By targeting P^* instead, profitability is preserved automatically across lot size changes - when lot rises, delta compresses proportionally; when lot falls, delta expands.

The ATRP(24h) signal scales P^* up and down with market conditions. In wide-range periods P^* expands, capturing available spread. In quiet periods P^* compresses, keeping sells close enough to fill quickly. Three guardrails bound the dynamic range: a floor at P^* equivalent to delta \$10 (where carry cost meets spread income), a ceiling at P^* equivalent to delta \$100 (where 24-hour fill rate drops below 95.8% on the actual tape), and asymmetric response speed - faster to widen in expanding range, slower to tighten in contracting range.

The base P^* is calibrated to the geometric midpoint of the valid parameter range in delta space: \$34.9, rounded to \$35. At a \$72 reference lot this implies P^* base = \$0.0255 per round trip. No discretion applied. The data produced the number.

4. **Perp futures hedge at $h=0.5$.** Sized to capture meaningful tail protection while limiting directional drag in benign markets. Recommended over $h=1.0$ based on the live backtest.
5. **Multi-bot parallel parameter testing.** Rather than backtest parameter sensitivity (subject to overfitting), run multiple instances with different lot sizes and deltas in parallel on live markets. 2x2 factorial design for static parameters in Phase 1; add ATRP-dynamic version in Phase 2; add hedge in Phase 3. Live A/B testing for genuine causal inference.
6. **Soft bag cap alarm.** Defense-in-depth against hedge failure. If cumulative bag exceeds 2x design-time peak, the bot pauses and notifies the operator. No automatic market-sells. Human decides next action.

What This Is, and What It Isn't

This is not a track record of edge. The strategy as run lost half of deployed capital. The decomposition shows the round-trip engine was working as designed and added latent value over the window; the realized losses came almost entirely from a capacity-relief mechanism that fired forced exits at unfavorable prices when capital and order-slot caps were hit simultaneously.

What this is: real reps, end-to-end ownership of a live trading system at non-trivial volume on a single venue, working risk controls, a clean and honest post-mortem, a falsified hypothesis (the perp hedge as primary fix) tested rigorously against the data, and a re-prioritized fix list with specific implementation steps.

I am looking for a trade operations or systematic-trading-adjacent seat at a serious shop where my research and operations background can pay its keep while I learn production trading from people who do it for real.

Period analyzed: Dec 2, 2024 to Jan 21, 2025. Single pair (BTC/USD). Single venue (Kraken). 336,644 fills. 99.99% passive maker. Self-funded. Self-built in Python.